

DORA YANG

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WORK EXPERIENCE

Sibel Health

Chicago, IL, USA

Systems Engineer II

Jun. 2025 - Present

Systems Engineer I

Sep. 2024 - Jun. 2025

- Project lead and sole systems engineer for a next-generation wireless pulse oximeter. Leading a 12-person cross-functional team (mechanical, electrical, firmware, algorithm, and manufacturing) through design reviews, scheduling, design transfer, and verification activities, including design history file management.
- Authored functional (vitals capture, data transmission) and performance (range, accuracy, battery life) system requirements, ensuring traceability and coverage across ~80 requirements and ~150 test cases.
- Built and maintained the risk management file by assessing probability and severity of hazards, mapping to and defining risk controls, and assessing residual risk after verification.

Hardware Test Engineer

Jan. 2024 - Sep. 2024

- Performed PCBA bring up and performance testing using bench equipment (power analyzers, oscilloscopes, multimeters, soldering station): power rail measurements, current consumption profiling, firmware flashing, and functional vital signs validation using calibrated simulators.
- Executed environmental verification testing by coordinating with external test labs for request for proposals, quotes/purchase orders, protocols, on-site supervision and report review.
- Designed and programmed an automated adhesive peel test fixture by sourcing and integrating a linear stage, optical encoders, and microcontroller, then mounting an off-the-shelf robotic arm and writing a Python control application reducing test time by 80% and eliminating manual operator involvement.

D-Wave Systems

Burnaby, BC, Canada

Electrical Engineer Co-op

May 2022 – Aug. 2022

- Redesigned the power supply for 20 quantum computers, replacing a faulty linear supply with a switch-mode alternative; built the full assembly including wiring, connectors, and a 3D-printed enclosure.
- Wrote STM32 firmware in C++ to bridge UART and RS-232 communication protocols between the PSU and control servers, and developed a Python GUI for setting output levels, and logging communication.
- Designed and built a resistive test load jig with power resistors, heat sinks, cooling fans, toggle switches, panel meters, and thermal fuses; reduced power supply build verification time by 30%.
- Developed a Python supply chain monitoring tool using the Octopart/Nexar API to query real-time stock availability for 2,100+ high-risk BOM components, presented findings to the supply chain team.

Precision Nanosystems Inc. (Now part of Cytiva)

Vancouver, BC, Canada

Instrumentation/Test Engineer Co-op

May 2021 – Dec. 2021

- Contributed to the development of a commercial microfluidic lipid nanoparticle formulation instrument (Cytiva NanoAssemblr®), characterizing fluidic system performance using flow, pressure, and temperature sensors to guide pump and materials selection.
- Built test jigs, authored protocols and reports, streamlined data processing and visualization by writing Python and MATLAB scripts for sensor measurement analysis, sampling distributions, and trend plots.

Innovation, Science and Economic Development (ISED) Canada

Ottawa, ON, Canada

Telecommunications Engineer Co-op

Jan. 2020 – Apr. 2020

- Assessed technical papers from telecommunications stakeholders for the deployment of 5G devices in Canada's 6GHz band, focusing on interference, device power, and safeguarding incumbent services.

EDUCATION

Bachelors of Applied Science in Engineering Physics

Sep. 2018 - May 2023

The University of British Columbia

Vancouver, BC, Canada

SKILLS

Software

Python, MATLAB, C++, Microsoft Office Suite (Excel, Word, Powerpoint), Jama, Matrix, Claude

Hardware

KiCAD, Altium, Soldering, Circuit Analysis, Oscilloscopes, Microcontrollers, Signal Generators

Certifications

Architecture of Complex Systems (MIT xPro), Models in Engineering (MIT xPro)

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PROJECTS

Flooded Cave Rescue Communication System

2023

- Designed and integrated a full-stack underwater wireless communication system, including flyback converter power electronics, H-bridge drive circuitry and firmware, achieving reliable text transmission at 100m range with a 2% error rate.
- Owned end-to-end system bring-up: reverse-engineered acoustic transducer behavior, iterated on analog front-end schematics (KiCAD), and validated signal integrity across the transmit/receive chain.
- Defined system-level performance targets (range, data rate, error rate) and verified them through structured bench testing, demonstrating the complete message path under simulated flooded cave conditions.

Electric Motor Dynamometer

2023

- Led a team of 3 students designing the hardware and software for a dynamometer to test UBC Supermileage's electric motor and powertrain systems.
- Designed and validated sensor signal conditioning circuits and an AC-to-DC converter for brake module control, then verified torque and RPM measurement accuracy against known loads.
- Built real-time data acquisition and display into the controller (Raspberry Pi Pico + LCD), enabling live performance monitoring and repeatable test execution across competition simulation runs.

CANBus Vehicle Accessory PCB

2022

- Designed a modular STM32-based PCB to interface vehicle accessories (lights, wipers, horn) over a CANBus network, defining the communication protocol and hardware-firmware interface for a hyper fuel efficient car.
- Owned schematic capture and PCB layout in KiCAD through to physical bring-up, debugging signal integrity and bus arbitration issues on live hardware.
- Authored and maintained firmware in C++ (Mbed) to handle CAN message parsing and accessory actuation, supporting reliable subsystem integration across the vehicle electrical architecture.

LIGO Coating Tester

2022

- Contributed to a precision measurement system for evaluating mechanical loss in optical coatings of gravitational wave detectors.
- Redesigning resonator disk geometry using COMSOL and ANSYS simulation to improve measurement sensitivity.
- Developed signal processing methods in Python to isolate coating loss signatures from noise, directly supporting experimental verification of material performance for gravitational wave interferometer components.

Recycling Robot

2021

- Designed and hand-built an autonomous robot integrating ultrasonic distance sensing, IR homing, H-bridge motor drive, and servo-actuated mechanisms for collecting aluminum cans, soldering all circuits and assembling the chassis.
- Implemented and debugged multi-mode firmware (C++) for line following, can detection, and bin homing on an STM32 platform, validating system behavior end-to-end against a timed competition objective.
- Characterized sensor performance and tuned control logic through iterative bench testing with a single-channel oscilloscope, demonstrating hands-on debug discipline under constrained tooling.